DATA LINEAGE REPORT

Used Car Price Prediction Pipeline — DS14260 Data Engineering

MILESTONE	INSTITUTION	DATASET	DATE
Milestone 3	PSUT (KSCS)	Kaggle S4E9	April 28, 2026

1. PIPELINE OVERVIEW

This document traces every transformation applied to the raw Kaggle dataset from initial ingestion through cleaning, feature engineering, modeling, and final submission output. The pipeline spans three project milestones and is implemented entirely in Python.

STAGE	MS	INPUT DATA	OUTPUT DATA	KEY OPERATIONS
Raw Ingestion	M1	train.csv, test.csv	DataFrames	`pd.read_csv`, schema check
Profiling	M1	Raw DataFrames	Profiling PDF, Issue Logs	Missingness, cardinality, IQR outlier scan
Cleaning	M2	Raw CSVs	train_cleaned, test_cleaned	Deduplication, lowercase, median impute, IQR winsorize
Feature Eng.	M2	Cleaned CSVs	train_features_v1, test_features_v1	car_age, engine_cc, OHE, StandardScaler, Yeo-Johnson
Baseline Model	M3	train_features_v1	Trained Model, Error Analysis	Log-price IQR filter, 80/20 train/val split, OLS Regression
Submission	M3	test_features_v1	final_predictions.csv	`model.predict` → `expm1()` target conversion

2. RAW DATA SOURCES

FILE	ROWS	COLS	HAS PRICE?	ORIGIN / DESCRIPTION
`train.csv`	188,533	13	Yes	Kaggle Playground Series S4E9 Training Set
`test.csv`	125,690	12	No	Kaggle Playground Series S4E9 Test Set
`sample_submission.csv`	125,690	2	No	Kaggle submission format template (id, price)

3. RAW COLUMN SCHEMA

COLUMN	DTYPE	PRESENT IN	ROLE	INITIAL LINEAGE / NOTES
`id`	int64	Train + Test	Identifier	Excluded from feature matrix; retained for submission join
`brand`	object	Train + Test	Categorical	57 unique brands; casing and whitespace noise present
`model`	object	Train + Test	Categorical	1,897 unique models; high-cardinality categorical
`model_year`	int64	Train + Test	Numeric	Vehicle manufacturing year; used to derive `car_age`
`milage`	int64	Train + Test	Numeric	Odometer reading in miles; right-skewed distribution
`fuel_type`	object	Train + Test	Categorical	7 categories; <5% missing values present
`engine`	object	Train + Test	Free-text	1,117 unique specs; parsed via regex to `engine_cc`
`price`	int64	Train only	Target	Continuous dollar amount; transformed via `log1p`

4. TRANSFORMATION PIPELINE DETAILS (M1 – M3)

Milestone 1 — Profiling: Generated schema dictionaries and issue logs capturing IQR outliers in `milage` and `price`, high-cardinality fields, and missing value counts across splits.

Milestone 2 — Cleaning & Feature Engineering: Applied explicit rules across features to prepare vectors for linear modeling:

FEATURE	CLEANING / IMPUTATION	DERIVED / ENGINEERED FEATURE	ENCODING & SCALING
`milage`	Median imputed (fitted on train); IQR winsorized	`milage_per_year` = milage / (car_age + 1)	StandardScaler → Yeo-Johnson
`model_year`	IQR winsorized [1886, 2024]	`car_age` = 2024 - model_year	StandardScaler → Yeo-Johnson
`engine`	Filled 'unknown'	`engine_cc` = regex extract displacement (L)	StandardScaler → Yeo-Johnson
`brand`, `model`	Stripped & lowercased	Categorical indicator columns	One-Hot Encoded (drop_first=True)
`price`	No imputation required	`target` = np.log1p(price)	Log-transformed target variable

5. MODEL TRAINING & EVALUATION (M3)

STAGE	METHOD / CONFIGURATION	PURPOSE & OPERATIONAL RESULT
Price Filter	1.5x IQR filter on `log1p(price)`	Removed extreme price outliers that distort OLS convergence
Train/Val Split	80/20 random split (seed = 42)	Validation set reserved for RMSE evaluation on real dollar values
Algorithm	Ordinary Least Squares (OLS) Linear Regression	Baseline linear model trained on scaled numerical and OHE features
Inversion	`np.expm1(y_pred)`	Converts log-space predictions back to actual currency values

6. OUTPUT ARTIFACTS INVENTORY

ARTIFACT NAME	FORMAT	PIPELINE ROLE
`train_data_dictionary.csv`	CSV	Milestone 1 data profiling metadata
`train_cleaned.csv` / `test_cleaned.csv`	CSV	Milestone 2 cleaned intermediate datasets
`train_features_v1.csv` / `test_features_v1.csv`	CSV	Milestone 2 model-ready feature matrices
`final_predictions.csv`	CSV	Milestone 3 final test set price predictions for Kaggle submission
`data_lineage_report.pdf`	PDF	Milestone 3 end-to-end data lineage documentation

> **Verification Notice:** This document represents the complete, fully anonymized data lineage report for the DS14260 Used Car Price Prediction project, verified for course submission and repository publication.